Data-Driven Development (D-D-D)

Let the stored data decide what the code should do next

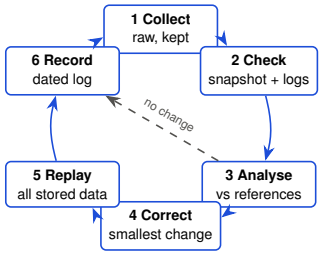
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Abstract. D-D-D is a simple development loop: *collect* the raw data, *check* it regularly, *analyse* what it shows against independent references, and *correct* the code only when the data say so — then *replay* the stored data through the new code and *record* what changed. Each step is ordinary; the method is in doing all of them, every time, with the raw data kept. This note describes the loop and how it has run during the development of MODE-S Wind, a research system that derives winds, temperatures and low-level windshear around Helsinki-Vantaa airport (EFHK) from passively received Mode S replies of aircraft. Seven worked cycles of October–November 2026 are summarised, including two in which the data said *not* to change the code.

1 Why another loop

In a system that measures the real world, most defects are not crashes but *plausible wrong numbers*: a wind 20 kt off, an alert that should not have been raised. Unit tests cannot find them, because nobody knows the right answer in advance; the only judge is the data itself, compared with something independent. Test-driven development writes the test before the code [1]; D-D-D takes its test cases from the stored data, after the fact, and keeps them. Its loop is the plan–do–check–act cycle of quality engineering [2], applied to code and measurements, with the replay of the stored data as the "check" — close in spirit to the controlled experiments of online software [3], but with yesterday's real traffic instead of an A/B split.

2 The loop



1 **Collect** — keep the raw inputs, not only the results: here every reply of every landing and departure is stored, so any later version of an algorithm can be run on any past flight.
2 **Check** — on a regular rhythm (here every few days and after each larger change) take a snapshot of the database and the logs: health, errors, gaps, the events of the period.
3 **Analyse** — compare with *independent* references: radiosondes, the MEPS weather model, a 327-m weather mast, METARs and pilot reports, and the other aircraft of the same hour. A finding is a repeatable difference, with numbers.
4 **Correct** — the smallest change that removes the finding. "No change" is a valid outcome.
5 **Replay** — every derived result carries a version; a new version is recomputed for all stored data in the background, so the change is measured before and after on the same flights, and on data it was not tuned on.
6 **Record** — a dated entry in a study log and the changelog: what was seen, what was changed, what the replay gave. The case that found the defect stays as a regression case.

3 Seven cycles in MODE-S Wind (October – November 2026)

Date	Finding in the data	Correction	Replay / result
10 Oct	Winds differ by airframe: 403 airframes, mean heading error +1.5°, repeatable (r = 0.91)	Per-airframe heading calibration, learnt online	Area wind against the sondes 11.5 → 9.1 kt RMS
19 Oct	Per-airframe temperature bias, SD 0.58 K	—	Out-of-sample gain 1 %: **not implemented**
27 Oct	Landings read 1.45 kt less, departures 1.38 kt more headwind than MEPS (16 228 bands)	Compare each flight phase with its own	Departure flags balanced: 19 / 16 instead of 27 / 9
27 Oct	Turboprop departures flagged 2.9 %, jets 0.6 %	Turboprop climb-out limits × 1.3	Turboprops 21 → 9 flagged; jets and landings unchanged
28 Oct	Short ramps of 11 kt reach the alert level	An alert needs a 14-kt change itself	4 155 landings: ramp alerts 3 → 1
4 Nov	False alert: an ATR with jammed GNSS reported a wrong ground speed	GNSS gate: such levels *unverified*	5 449 landings: 2 alerts unverified; the real frontal alerts unchanged
2–4 Nov	Mast wind directions off by +15° / −9° in every wind sector	— (the *reference* is wrong)	**No change**: raw values kept, documented

One cycle, one day: 4 November 2026

The routine check of the day's snapshot listed the windshear events of the last day and a half. One alert of the evening before did not fit: an ATR 72 on final read 59 kt of wind at 1 200–1 400 ft while every aircraft before and after it read 27 kt. Its own data told why — position accuracy zero, a frozen position, no ADS-B velocity, and a ground speed 20–30 kt below what its airspeed and the wind allowed: GNSS jamming. The correction (a level at heights where the GNSS was degraded and no ADS-B velocity was sent becomes *unverified*, not an alert) was first tried on all 5 449 stored landings: two reports changed, both false, while the real alerts of a frontal passage the next morning stayed. A first, wider rule was rejected by the same replay because it also silenced two jets whose winds were good. Check, analysis, correction, replay and record took a few hours.

4 What makes it work

- **Raw data are the asset.** Without the stored replies none of the replays above would have been possible; storage was budgeted for it from the start (about 12 MB a day).
- **Versioned results.** A report version in every stored result makes "recompute everything" a routine start-up job, not a migration project.
- **More than one reference.** A difference that the model, the sonde and the aircraft share points at the reference (the mast case); one that only the aircraft show points at the code.
- **Waiting is allowed.** When a finding rests on a handful of cases — three ATR departures in early November — the decision is to collect more, with a date to look again.
- **Small, reversible steps**, each judged by one replay.
- **Fast turnaround.** A snapshot, an analysis and a replay of all landings take minutes; the loop runs in hours, not release cycles.

5 Limits

D-D-D finds what the data and the references can show: an error the references share, or a case that never occurred, stays hidden. It needs storage, discipline in the checks, and honesty in the record — a "no change" result written down is as valuable as a fix. MODE-S Wind is research only; nothing in it is used for flight operations.

References

[1] Beck, K. (2002). *Test-Driven Development: By Example*. Addison-Wesley.
[2] Deming, W.E. (1986). *Out of the Crisis*. MIT Press (the plan–do–check–act cycle).
[3] Kohavi, R., Tang, D., Xu, Y. (2020). *Trustworthy Online Controlled Experiments*. Cambridge University Press.